Script-Controlled ACL – Restrict Record Access Based on Field Value

Project Design Phase: Proposed Solution Template

Date:	October 3, 2026	**Maximum Marks:**	4 Marks
Team ID:	SWTID-2026-8089	**Project Name:**	Script-Controlled ACL

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No	Parameter	Description
1.	Problem Statement (Problem to be solved)	In ServiceNow, sensitive records such as student and institution details are exposed to any user who has basic table access. Role-only security cannot restrict visibility by field value (for example, the Branch field), so users from one branch can view, edit, create or delete records belonging to another branch. This causes data privacy risks, unauthorized modifications, accidental deletions and non-compliance with record-level security requirements.
2.	Idea / Solution description	A record-level security solution built on ServiceNow using script-controlled Access Control Lists (ACLs) on a custom table, Institution Details (u_institution_details). Custom roles (bb1, bb2, bb3, bb4) are assigned to a test user (EEE User). A READ ACL combines the role check, the data condition (Branch is EEE) and an advanced script (gs.hasRole) so that only bb1 users see EEE branch records while admins retain full access. Separate CREATE (bb2), WRITE (bb3) and DELETE (bb4) ACLs control who can add, edit and remove records. Behaviour is verified by impersonating the EEE user and the admin.
3.	Novelty / Uniqueness	• Script-controlled ACLs that evaluate user roles and record field values together for each access decision. • Granular, operation-wise separation of duties: Read (bb1), Create (bb2), Write (bb3) and Delete (bb4) through dedicated roles. • Admin override retained through script logic, so administrators keep full access while all other users are denied by default. • Fully configuration-based and reusable on any table without changing application code.

S.No	Parameter	Description
4.	Social Impact / Customer Satisfaction	Customer Satisfaction: Gives institutions confidence that student data is visible only to authorized users, improving trust and transparency. Data Privacy and Security: Protects sensitive records from unauthorized viewing, modification, creation and deletion across forms, lists and Service Portal interfaces. Operational Efficiency: Removes the need for manual data segregation and reduces the risk of data-handling errors.
5.	Business Model (Revenue Model)	Internal Enterprise Value: Lowers the cost of data breaches and audit failures by enforcing least-privilege access, and reduces administrative effort spent on manual access reviews. Commercial Consulting / Template Model: A reusable ACL security template and implementation package that can be offered to educational institutions and enterprises that need ServiceNow record-level security.
6.	Scalability of the Solution	Highly scalable across any organization size. The same ACL pattern can be applied to additional tables and fields (such as Department, Location or Region), extended with new roles and data conditions for other branches (ECE, CSE), and adapted for Service Portal and multi-department use without redesigning the core security logic.